CV of the researcher



• Personal information

Name: Rodríguez-Mayorga, Mauricio Antonio
Nationality: Spanish
ORCID: http://orcid.org/0000-0003-0869-9787
Research gate: https://www.researchgate.net/profile/Mauricio_Mayorga2
GitHub: https://github.com/marm314
Web site: https://marm314.github.io/index.html
email: marm3.14@gmail.com
h-index: 15, h10-index: 20

• Education

Jun. 26.06.2023 Master in Teaching in High-School, Baccalaureate, Vocational Training, and languages (**specialized in teaching Chemistry and Physics**), Univ. of Sevilla (Spain).

Jul. 24.07.2018 Ph.D. in Chemistry, Univ. Girona (Spain). Excellent Cum laude. Under the supervision of Prof. M. Solà and Dr. E. Matito

Sep. 06.09.2013 Master in Theoretical and Computational Chemistry, Univ. Rovira i Virgili (Spain). **(Graduated first in my class)**

Jul. 30.06.2012 BSc. in Chemistry, Univ. Granada (Spain). **(Graduated first in my class)**

Summer schools attended:

3) Title: 2nd edition of the International summer School in electronic structure Theory: electron correlation in Physics and Chemistry (ISTPC). Organizer: University of Strasbourg. Place: Centre Paul Langevin, Aussois, Savoie, France. Date: 01/07/2017. Type: International. Days: 13.

2) Title: Hands-on workshop density-functional theory and beyond: First principles simulations of molecules and materials. Organizer: Fritz-Haber-Institut der Max-Planck-Gesellschaft. Place: Berlin, Germany. Date: 23/07/2015. Type: International. Days: 11.

1) Title: Sostrup Summer School - Quantum Chemistry and Molecular Properties. Organizer: University of Aarhus. Place: Ry, Denmark. Date: 18/07/2014. Type: International. Days: 13.

• Employment history including current position/s **with the name/s of the advisor/s**

01.01.2026 **Postdoc researcher at Prof. P.-F. Loos and Dr. C. Mejuto-Zaera group, Laboratoire de Chimie et Physique Quantiques and Laboratoire de Physique** (current position) **Théorique - Toulouse (France)**

01.01.2025 - 31/12/2025 Postdoc researcher at Prof. P.-F. Loos group, Laboratoire de Chimie et Physique Quantiques - Toulouse (France)

01.07.2023 - 31/12/2024 Postdoc researcher at Profs. G. D'Avino and X. Blase group, Institute Néel Grenoble (France)

01.02.2023 - 31.06.2023 Postdoc researcher at Prof. J. C. Sancho-García's group, Universitat d'Alacant (Spain)

15.01.2021 - 31.12.2022 Postdoc researcher at Prof. L. Visscher's group, Vrjie Universiteit Amsterdam(Netherlands) with a Marie-Curie individual fellowship

01.01.2020 - 31.12.2020 Postdoc researcher at Dr. F. Bruneval's group, Univ. of Paris Saclay (France)

01.06.2019 - 30.11.2019 Postdoc researcher at Prof. S. Luber's group, Univ. of Zürich (Switzerland)

15.09.2013 - 24.07.2018 Ph.D. student, Univ. of Girona (Spain).

01.07.2012 - 06.09.2013 Research assistant, Univ. of Girona (Spain).

• Approved research projects

I have participated in three research projects from the Spanish ministry and the European Union:

6. Title: RAtional Pathways TOwards high performance ORganic Semiconductors - RAPTORS
Funding agency: Agence National de la Recherche - France.
PI: Dr. Gabriele D'Avino. Amount awarded: 298 136 €

5. Title: Relativistic reduced density matrix functional theory
Funding agency: **Marie Sklodowska-Curie Individual Fellowship**
PI: Dr. Mauricio Rodríguez-Mayorga. Amount awarded: 180 000 €

4. Title: GW reduced density matrix for solids. Implementation in ABINIT.
Funding agency: CEA Cross cutting program "Numerical simulations".
PIs: Dr. Fabien Bruneval and Dr. Marc Torrent. Amount awarded: 56 000 €

3. Title: DFT Functionals for the computation of non-linear optical properties
Funding agency: MINECO Ref CTQ2014-52525-P, Spanish government
PIs: Dr. Eduard Matito and Dr. Josep Maria Luis. Amount awarded: 75 000 €

2. Title: Development of new non-empirical DFT functionals
Funding agency: FP7-PEOPLE-2011-CIG Marie-Curie Action: "Career Integration Grants" European Council.
PI: Dr. Eduard Matito. Amount awarded: 100 000 €.

1. Title: A new stragey in DFT: hybrids functionals adapted to electron correlation
Funding Agency: Programa Europa Excelencia, Spanish government
PI: Dr. Eduard Matito. Amount awarded: 75 000 €

• Tutoring junior researchers (summary information, the names of the junior researchers should be indicated)

5) Mentoring: Final degree project of Mr. D. Këizer (2022). Center: Vrije Universiteit Amsterdam (Netherlands).

4) Mentoring: Final master project of Mr. C. Naim (2018). Center: Donostia International Physics Center (Spain)

3) Mentoring: Final master project of Mr. S. Sitkiewicz (2017). Center: University of the Basque Country (Spain)

2) Mentoring: Summer interships at DIPC for undergraduate students: Mr. Martí Gimferrer (2017), Mr. Carmelo Naim (2017), and Mr. Juan Reino (2016). Center: Donostia International Physics Center (Spain).

1) Mentoring: Final degree project of Ms. N. Abullí (2012). Center: University of Girona (Spain). This project was awarded with the extraordinary degree award.

• Teaching activities (summary information)

5) Subject: Physics and Chemistry to 2nd year High-school students. Center: I.E.S. Chaves Nogales (Sevilla). Credits: 10 ECTS, Dates: March 2023. [As part of the training of the master of education]

4) Subject: Integrated scientific techniques II. Center: University of Girona (Spain). Credits: 2.5 ECTS. Dates: January 2017.

3) Subject: Principles of chemical reactivity. Center: University of Girona (Spain). Credits: 4 ECTS. Dates: 26/09/16 to 30/09/2016

2) Subject: Principles of chemical reactivity. Center: University of Girona (Spain). Credits: 9 ECTS. Dates: 29/2/16 to 18/3/2016

1) Subject: Integrated scientific techniques II. Center: University of Girona (Spain). Credits: 5.5 ECTS. Dates: 10/11/2015 to 27/11/15

• Memberships in panels, boards, etc., and individual scientific reviewing activities

- Reviewer for J. Chem. Theory Comput., J. Chem. Phys., Nature Communications, and MDPI.

• Prizes, awards, fellowships

Jan. 2021 - Dec. 2022 ‘Marie Sklodowska-Curie individual fellowship’
Project: 891647-ReRDFMT

Aug. 2014 - Dec. 2017 ‘Beca de formación de Personal Universitario (FPU)’
(PhD grant, Spanish Goverment)

Feb. 2014 - Sep. 2014 'Beca de formació d'investigadors (FI)'
(PhD grant, Catalan government)- declined in favor to the FPU

Jul. 2013 Awarded with the Masters' Extraordinary Degree award from the University Rovira I Virgili (Spain).

Jul. 2012 Awarded with the Bachelors' Extraordinary Degree award from the University of Granada (Spain).

Jan. 2012 - Jun. 2012 'Beca Colaboración' (Highly competitive undergraduate fellowship form the Spanish Goverment for doing research at bachelor level)

• Personal skills (e.g. language skills, digital competences)

Spanish: native (mother tongue), English: fluent, and French: intermediate.

Good programming skills in Python, Fortran (F90) and C++.

Experience in Parallel programming (MPI and Open-MPI) and on building servers

Major scientific achievements

During my Ph.D. I worked under the supervision of Dr. E. Matito in collaboration with the group of Prof. J.M. Ugalde[1] using computational chemistry and **developing tools** (for the analysis of the intracule and extracule probability densities) that were used to interpreted the results obtained using computational chemistry methods/packages. To that end, **I developed some computational chemistry packages** that are available and can be downloaded from GitHub (RHO_OPS [https://github.com/marm314/rho_ops] and RHO2_OPS [https://github.com/marm314/rho2_ops]). I also collaborated with Prof. M. Piris[1] in the development of reduced density matrix functional approximations. I designed the following benchmark, which I signed as **corresponding author (during my PhD)**

- Mitxelena I., Piris M., Rodríguez-Mayorga M.; On the performance of Natural Orbital Functional Approximations in Hubbard Model. *J. Phys. Condens. Matter.,* **29**, 425602 (2017) doi:10.1088/1361-648X/aa80ca ***Article selected by the journal among the Highlights of the 2017***

Later, as postdoc, in colaboration with Dr. F. Bruneval[2], I suggested: a) a method to improve the functionals used in reduced density matrix functional theory (RDMFT) by including many-body perturbation theory corrections

- Rodríguez-Mayorga M., Mitxelena I., Bruneval F., Piris M.; .Coupling Natural Orbital Functional Theory and Many-Body Perturbation Theory by Using Nondynamically Correlated Canonical Orbitals **J. Chem. Theory Comput.**, 17, 7562 (**2021**) doi.org/10.1021/acs.jctc.1c00858

1 Donostia International Physics Center (DIPC) in Donostia-San Sebastian (Spain)

2 Université Paris Saclay (France)

and b) the usage of the random-phase approximation(RPA) for computing non-linear optical properties (i.e. (hyper)polarizabilities)

- Besalú-Sala, P., Bruneval, F., Pérez-Jiménez, Á. J., Sancho-García, J. C., Rodríguez-Mayorga, M.; RPA, an accurate and fast method for the computation of non-linear optical properties. **J. Chem. Theory Comput. (2023)** doi.org/10.1021/acs.jctc.3c00674

I signed both works as **corresponding author.**

As Marie Curie Fellow, **I proposed the extension of the applicability of RDMFT to the relativistic framework** in collaboration with Dr. K.J.H. Giesbertz and Prof. L. Visscher[3]

- Rodríguez-Mayorga M., Giesbertz K.J.H., Visscher L.; Relativistic reduced density matrix functional theory **SciPost Chem.**, 1, 004 (**2022**) doi:10.21468/SciPostChem.1.2.004

Finally, I have **participated in 23 national and international conferences** with oral contributions in 14 (2 as an invited speaker) and poster presentations in 9.

• Full list of publications

Papers signed as the corresponding author, paper signed as first author.

30) Johnson P.A., Fecteau C.-E., Nadeau S., Rodriguez-Mayorga M., Loos P.-F.; Connections between Richardson–Gaudin states, perfect-pairing, and pair coupled-cluster theory **J. Chem. Phys.**, 163, 234115 (**2025**) doi.org/10.1063/5.0305612
IF: 3.10

29) Verstraete M. J., Abreu J., Allemand G. E., Amadon B., Antonius G., Azizi M., Baguet L., Barat C., Bastogne L., Bejaud R., Beuken J.-M., Bieder J., Blanchet A., Bottin F., Bouchet J., Bouquiaux J., Bousquet E., Boust J., Brieuc F., Brousseau-Couture V., Brouwer N., Bruneval F., Castellano A., Castiel E., Charraud J.-B., Clerouin J., Cote M., Duval C., Gallo A., Gendron F., Geneste G., Ghosez P., Giantomassi M., Gingras O., Gomez-Ortiz F., Gonze X., Goudreault F. A., Gruneis A., Gupta R., Guster B., Hamann D. R., He X., Hellman O., Holzwarth N., Jollet F., Kestener P., Lygatsika I.-M., Nadeau O., MacEnulty L., Marazzi E., Mignolet M., O'Regan D. D., Outerovitch R., Paillard C., Petretto G., Ponce S., Ricci F., Rignanese G.-M., Rodriguez-Mayorga M., Romero A. H., Rostami S., Royo M., Sarraute M., Sasani A., Soubiran F., Stengel M., Tantardini C., Torrent M., Trinquet V., Vasilchencko V., Waroquiers D., Zabalo A., Zadoks A., Zhang H., Zwanziger J.; Abinit 2025: New Capabilities for Predictive Modeling of Solids and Nanomaterials **J. Chem. Phys.**, 163, 164126 (**2025**) doi.org/10.1063/5.0288278
IF: 3.10

28) Rodríguez-Mayorga M., Loos P.-F., Bruneval F., Visscher L.; Time-reversal symmetry in RDMFT and pCCD with complex-valued orbitals **J. Chem. Phys.**, 162, 054716 (**2025**) doi.org/10.1063/5.0242504
IF: 3.10

27) Rodríguez-Mayorga M., Blase X., Duchemin I., D'Avino G.; From many-body ab initio to effective excitonic models: a versatile mapping approach including environmental embedding effects **J. Chem. Theory Comput.**, 20, 8675 (**2024**) doi.org/10.1021/acs.jctc.4c00756

3 Vrjie Universiteit Amsterdam (The Netherlands)

IF: 5.50

26) Ammar A., Marie A., Rodríguez-Mayorga M., Burton H., Loos P.-F.; Can *GW* Handle Multireference Systems? **J. Chem. Phys.**, 160, 114101 (**2024**) doi.org/10.1063/5.0196561
IF: 3.10

25) Rodríguez-Mayorga M., Besalú-Sala P., Pérez-Jiménez A.J., Sancho-García J.C.; Application to nonlinear optical properties of the RSX-QIDH double-hybrid range-separated functional. **J. Comput. Chem.**, 1, 1 (**2024**) doi.org/10.1002/jcc.27302
IF: 4.80

24) Bremond, E., Rodríguez-Mayorga M., F., Pérez-Jiménez, Á. J., Adamo, C., Sancho-García; Assessment of the nonempirical r^2SCAN-QIDH double-hybrid density functional against large and diverse datasets. **J. Chem. Phys.,** 159, 141101 **(2023)** doi.org/10.1063/5.0157259

IF: 3.10

23) Besalú-Sala, P., Bruneval, F., Pérez-Jiménez, Á. J., Sancho-García, J. C., Rodríguez-Mayorga, M.; RPA, an accurate and fast method for the computation of non-linear optical properties. **J. Chem. Theory Comput.** 19, 6062 **(2023)** doi.org/10.1021/acs.jctc.3c00674

IF: 5.50

22) Hassan-Denawi, Bruneval. F., A., Torrent. M., Rodríguez-Mayorga, M.; *GW* density matrix to estimate self-consistent *GW* total energy in solids. **Phys Rev. B.**, 108, 125107 **(2023)** doi.org/10.1103/PhysRevB.108.125107

IF: 3.60

21) Rodríguez-Mayorga M.; Standalone NOFT module (1.0). **Zenodo** (**2022**). doi.org/10.5281/zenodo.735273

20) Rodríguez-Mayorga M., Keizer D., Giesbertz K.J.H., Visscher L.; Relativistic effects on electronic pair densities: a perspective from the radial intracule and extracule probability densities. **J. Chem. Phys.**, 157, 194301 (**2022**) doi.org/10.1063/5.0127190

IF: 4.40

19) Telleria X., Ugalde J.M., Lopez X., Matito E., Ramos-Cordoba E., Rodríguez-Mayorga M.; Aufbau Principle and Singlet-Triplet Gap in Spherical Hooke Atoms **Int. J. Quantum Chem.**, e27019 (**2022**) doi.org/10.1002/qua.27019

IF: 2.20

18) Rodríguez-Mayorga M., Giesbertz K.J.H., Visscher L.; Relativistic reduced density matrix functional theory **SciPost Chem.**, 1, 004 (**2022**) doi:10.21468/SciPostChem.1.2.004

IF: 3.20

17) Via-Nadal M., Rodríguez-Mayorga M., Ramos-Cordoba E., Matito E.; Natural Range Separation of the Coulomb Hole **J. Chem. Phys.**, 156, 184106 (**2022**) doi.org/10.1063/5.0085284

IF: 4.39

16) Rodríguez-Mayorga M., Mitxelena I., Bruneval F., Piris M.; .Coupling Natural Orbital Functional Theory and Many-Body Perturbation Theory by Using Nondynamically Correlated Canonical Orbitals **J. Chem. Theory Comput.**, 17, 7562 (**2021**) doi.org/10.1021/acs.jctc.1c00858

IF: 6.21

15) Bruneval F., Rodríguez-Mayorga M., Rinke P., Dvorak M.; Improved One-Shot Total Energies from the Linearized *GW* Density Matrix. **J. Chem. Theory Comput.**, 17, 2126 (**2021**) doi.org/10.1021/acs.jctc.0c01264

IF: 6.21

14) Han R., Rodríguez-Mayorga M., Luber S.; A Machine Learning Approach for MP2 Correlation Energies and Its Application to Organic Compounds. **J. Chem. Theory Comput.**, 17, 777 (**2021**) doi.org/10.1021/acs.jctc.0c00898

IF: 6.21

13) Via-Nadal M., Rodríguez-Mayorga M., Ramos-Cordoba E., Matito E.; Singling Out Dynamic and Nondynamic Correlation. **J. Phys. Chem. Lett.**, 10, 4032 (**2019**) doi.org/10.1021/acs.jpclett.9b01376

Article awarded with the GEQC-RSEQ 2021 Methodology Development Prize

IF: 6.71

12) Sitkiewicz S., Rodríguez-Mayorga M., Luis J.M. , Matito E.; Partition of Optical Properties into Orbital Contributions. **Phys. Chem. Chem. Phys.**, 21, 15380 (**2019**) doi.org/10.1039/C9CP02662B

Article selected by the journal as HOT PCCP article

IF: 3.68

11) Quintero-Monsebaiz, R., Mitxelena I., Rodríguez-Mayorga, M., Vela, A., and, Piris, M.; Natural orbital functional for spin-polarized periodic systems. **J. Phys. Condens. Matter.**, 31, 165501 (**2019**) doi.org/10.1088/1361-648X/ab0170

IF: 2.33

10) Rodríguez-Mayorga M., Ramos-Cordoba E., Lopez X., Solà M., Ugalde J. M., Matito E.; The Coulomb Hole of the Ne Atom. **Chemistry Open** ,8, 1 (**2019**) doi:10.1002/open.201800235 ***cover***

IF: 2.37

9) Mitxelena I., Rodríguez-Mayorga M., Piris M.; Phase Dilemma in Natural Orbital Functional Theory from the N-representability Perspective. **Eur. Phys. J. B** 91, 109 (**2018**) doi:10.1140/epjb/e2018-90078-8

IF: 1.35

8) Rodríguez-Mayorga M., Via-Nadal M., Solà M., Ugalde J. M., Lopez X., Matito E.; Electron-Pair Distribution in Chemical Bond Formation. **J. Phys. Chem. A** 122, 1916 (**2018**) doi:10.1021/acs.jpca.7b12556

IF: 2.64

7) Via-Nadal M., Rodríguez-Mayorga M., Matito E.; A Salient Signature of van der Waals interactions. **Phys. Rev. A rapid comm.** 96, 050501(R)(**2017**) doi:10.1103/PhysRevA.96.050501

IF: 3.99

6) Rodríguez-Mayorga M., Ramos-Cordoba E., Via-Nadal M., Piris M., Matito E.; Comprehensive Benchmarking of Density Matrix Functional Approximations. **Phys. Chem. Chem. Phys.** 19, 24029 **(2017)** doi:10.1039/C7CP03349D

IF: 3.91

5) Mitxelena I., Piris M., Rodríguez-Mayorga M.; On the performance of Natural Orbital Functional Approximations in Hubbard Model. **J. Phys. Condens. Matter.** 29, 425602 **(2017)** doi:10.1088/1361-648X/aa80ca

Corrigendum: **J. Phys. Condens. Matter.** 30, 089501 **(2018)** doi:10.1088/1361-648X/aaa659

Article selected by the journal among the Highlights of the 2017

IF: 2.62

4) Rodríguez-Mayorga M., Ramos-Cordoba E., Feixas F., Matito E.; Electron Correlation Effects in Third-Order Densities. **Phys. Chem. Chem. Phys.** 19, 4522 (**2017**) doi:10.1039/C6CP07616E

IF: 3.91

3) Mercero J.M., Mayorga-Rodríguez M., Matito E., Lopez X., Ugalde J.; The Electron-pair Density Distribution of the $^{1,3}\Pi_u$ Excited States of H_2. **Can. J. Chem.** 94, 998 (**2016**) doi:10.1139/cjc-2016-0203

IF: 1.08

2) Rodríguez-Mayorga M., Ramos-Cordoba E., Salvador P., Solà M., Matito. E.; Bonding description of the Harpoon Mechanism. **Mol. Phys.** 114, 1345 (**2016**) doi:10.1080/00268976.2015.1121297

IF: 1.87

1) Feixas F., Rodríguez-Mayorga M., Matito E., Solà M.; Three-center bonding analyzed from correlated and uncorrelated third-order reduced density matrices. **Comp. Theor. Chem.** 1053, 173(**2015**) doi:10.1016/j.comptc.2014.09.030

IF: 1.47